Regression: Model Fit and Interpretation

Week 3, Day 1 and Day 2

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1 What is Regression?

Regression is a statistical method to **model and analyze the relationship** between:

- a **dependent variable** (response), here `yield`
- one or more **independent variables** (predictors), here `nitrogen` and `rainfall`

Intuition: we draw the "best" line (or plane) through the data, so we can say *how much yield changes when nitrogen or rainfall changes*, and *predict yield* for new fields.

Units used in this file: yield and soil nitrogen in **kg/ha**, rainfall in **mm**.

```
library(ggplot2)
library(knitr)
```

2 Day 1: Building the Models

2.1 Read the data

```
data <- read.csv("../data/raw/agriculture_yield_data.csv")
head(data)
```

```
  nitrogen rainfall yield
1    49.49   180.91 4.807
2    37.20   165.36 4.189
3    63.24   201.93 5.860
4    64.64   208.05 6.231
5    54.23   190.15 5.204
6    60.91   197.76 5.884
```

2.2 Fit two models

```r
# Model 1: one predictor
model  <- lm(yield ~ nitrogen, data = data)

# Model 2: two predictors
model2 <- lm(yield ~ nitrogen + rainfall, data = data)
```

- `model` asks: *can nitrogen alone predict yield?*
- `model2` asks: *does adding rainfall help?*

2.3 Model summary

```r
summary(model)
```

```
Call:
lm(formula = yield ~ nitrogen, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.37915 -0.06810  0.01932  0.12831  0.22899

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 1.517060   0.362919    4.18  0.00308 **
nitrogen    0.069383   0.005967   11.63 2.72e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.188 on 8 degrees of freedom
Multiple R-squared:  0.9441,    Adjusted R-squared:  0.9372
F-statistic: 135.2 on 1 and 8 DF,  p-value: 2.725e-06
```

```r
summary(model2)
```

```
Call:
lm(formula = yield ~ nitrogen + rainfall, data = data)

Residuals:
```

```
     Min       1Q   Median       3Q      Max
-0.33849 -0.07160 -0.00697  0.11169  0.22162

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.40266    2.13631  -0.188    0.856
nitrogen     0.03798    0.03495   1.087    0.313
rainfall     0.01902    0.02085   0.912    0.392

Residual standard error: 0.19 on 7 degrees of freedom
Multiple R-squared:  0.9501,    Adjusted R-squared:  0.9358
F-statistic:  66.6 on 2 and 7 DF,  p-value: 2.781e-05
```

```
s <- summary(model2)
```

3 Reading summary(model2)

The model:

```
lm(formula = yield ~ nitrogen + rainfall, data = data)
```

It tries to predict crop yield using two predictors: nitrogen and rainfall.

3.1 Coefficients table

Term	Estimate	Std. Error	t value	p value	Meaning
Intercept	-0.40266	2.13631	-0.188	0.856	When nitrogen = 0 and rainfall = 0, predicted yield = -0.40266 kg/ha. This is not realistic (used for calculation only), and not statistically significant.
Nitrogen	0.03798	0.03495	1.087	0.313	For every 1 unit increase in nitrogen, yield increases by 0.03798 kg/ha (not statistically significant).
Rainfall	0.01902	0.02085	0.912	0.392	For every 1 mm increase in rainfall, yield increases by 0.01902 kg/ha (not statistically significant).

How to interpret p-values:

- $p < 0.05$: statistically significant (important)
- $p >= 0.05$: not significant on its own

If none of the variables is significant individually but the model is strong overall, the predictors

probably **overlap** (they carry similar information). This is why 03 studies correlation and multicollinearity.

3.2 Residuals

Residual = actual yield - predicted yield. It is how far each prediction is from the truth.

```
res <- residuals(model2)
round(quantile(res), 4)
```

```
     0%     25%     50%     75%    100%
-0.3385 -0.0716 -0.0070  0.1117  0.2216
```

- Smaller values = better prediction
- Symmetric and small spread (median near 0) = good sign, no major outliers

3.3 Residual standard error (RSE)

```
rse <- s$sigma
round(rse, 4)
```

```
[1] 0.19
```

- The typical distance between actual and predicted yield, **in yield units**
- Lower = better
- Compare it with the average yield to judge if it is small:

```
round(rse / mean(data$yield) * 100, 1)
```

```
[1] 3.3
```

The number above is RSE as a percentage of mean yield.

3.4 Multiple R-squared

```r
round(s$r.squared, 4)
```

```
[1] 0.9501
```

- Share of yield variation explained by nitrogen + rainfall
- Close to 1 = strong fit
- Warning: R^2 goes up even when you add **useless** variables

3.5 Adjusted R-squared

```r
round(s$adj.r.squared, 4)
```

```
[1] 0.9358
```

- R^2 adjusted for the number of variables
- Better than R^2 when comparing models with different numbers of predictors
- If adjusted R^2 is still high, the fit is realistic, not just inflated by extra variables

3.6 F-statistic

```r
fs   <- s$fstatistic
f_p  <- pf(fs[1], fs[2], fs[3], lower.tail = FALSE)
c(F = unname(fs[1]), df1 = unname(fs[2]), df2 = unname(fs[3]), p_value = unname(f_p))
```

```
           F          df1          df2      p_value
6.659873e+01 2.000000e+00 7.000000e+00 2.781331e-05
```

- Tests: *do nitrogen and rainfall together explain yield variation?*
- High F and small p (< 0.001) = the model is significant overall
- A model can be significant overall even if individual variables are not (overlapping predictors)

3.7 Final evaluation (what is good?)

```r
eval_tbl <- data.frame(
  Metric = c("R-squared", "Adjusted R-squared", "Residual std. error",
             "F-statistic p-value", "Variable p-values"),
  Value = c(round(s$r.squared, 4),
            round(s$adj.r.squared, 4),
            round(rse, 4),
            signif(f_p, 3),
            paste(round(coef(s)[-1, 4], 3), collapse = ", ")),
  Verdict = c(ifelse(s$r.squared > 0.7, "Strong fit", "Weak or moderate fit"),
              ifelse(s$adj.r.squared > 0.7, "Realistic fit", "Weak or moderate fit"),
              "Small relative to yield = good",
              ifelse(f_p < 0.05, "Very significant", "Not significant"),
              ifelse(all(coef(s)[-1, 4] < 0.05), "All significant",
                     "Not all significant alone"))
)
kable(eval_tbl)
```

Metric	Value	Verdict
R-squared	0.9501	Strong fit
Adjusted R-squared	0.9358	Realistic fit
Residual std. error	0.19	Small relative to yield = good
F-statistic p-value	2.78e-05	Very significant
Variable p-values	0.313, 0.392	Not all significant alone

4 How R-squared Relates to Prediction

4.1 Actual, predicted and mean

- **Actual** = the real observed value you measured. Example: a field gave 3.5 t, so actual yield = 3.5.
- **Predicted** = what the model estimated from nitrogen, rainfall, etc. Example: the model said 3.2 t.
- **Mean** = average of all actual yields (add all, divide by the number of observations). It is the "center" of the yield values.

```r
actual    <- data$yield
predicted <- fitted(model2)
m         <- mean(actual)
```

```r
head(data.frame(actual = actual, predicted = round(predicted, 3),
                residual = round(actual - predicted, 3)))
```

```
  actual predicted residual
1  4.807     4.918   -0.111
2  4.189     4.155    0.034
3  5.860     5.840    0.020
4  6.231     6.009    0.222
5  5.204     5.274   -0.070
6  5.884     5.672    0.212
```

4.2 Three types of variation

Type	Formula	Meaning
Total variation	SST = sum(Actual - Mean)^2	how much yield varies overall
Explained variation	SSR = sum(Predicted - Mean)^2	how much the model explains
Unexplained variation	SSE = sum(Actual - Predicted)^2	error left by the model

Intuition: SST is the total "spread" of yield. The model splits it into a part it can explain (SSR) and a part it cannot (SSE). So SST = SSR + SSE.

```r
SST <- sum((actual - m)^2)
SSR <- sum((predicted - m)^2)
SSE <- sum((actual - predicted)^2)

kable(data.frame(Measure = c("SST", "SSR", "SSE", "SSR + SSE"),
                 Value = round(c(SST, SSR, SSE, SSR + SSE), 4)))
```

Measure	Value
SST	5.0598
SSR	4.8072
SSE	0.2526
SSR + SSE	5.0598

4.3 How R-squared is calculated

R^2 = Explained variation / Total variation = SSR / SST

```
R2_manual <- SSR / SST
c(manual = R2_manual, from_summary = s$r.squared)
```

```
    manual from_summary
 0.9500704    0.9500704
```

The two values match. So:

- $R^2 = 0.95$ means 95% of the total variation in yield is explained by nitrogen and rainfall.
- The remaining 5% is error (unexplained).

4.4 R-squared is not "accuracy"

$R^2 = 0.95$ means the model **explains** 95% of the yield variation in this data. It does **not** mean the model is "95% accurate", and it does not guarantee future prediction accuracy if important factors are missing.

5 Overfitting

- Happens when the model uses **too many variables**, including ones that are not really important.
- It performs well on the current data but poorly on new data (it learns the noise, not the pattern).

Why R^2 alone can be misleading: a high R^2 could mean overfitting. That is why **adjusted R^2** is preferred: it goes down when unnecessary variables are added.

Read it like this:

- R^2 never goes down when a variable is added, even a useless one.
- Adjusted R^2 drops (or barely rises) for a useless variable. Trust adjusted R^2 when comparing models.

6 What if Other Important Factors are Missing?

- Example: **irrigation** may be important but is not in the model.
- The model then assumes conditions like irrigation are **constant or optimal**.
- If that is not true, predictions may be **biased**.

So a good R^2 only speaks for the variables you included. Always ask: *what else affects yield that I did not measure?*

7 Day 2: Visualizing the Relationship

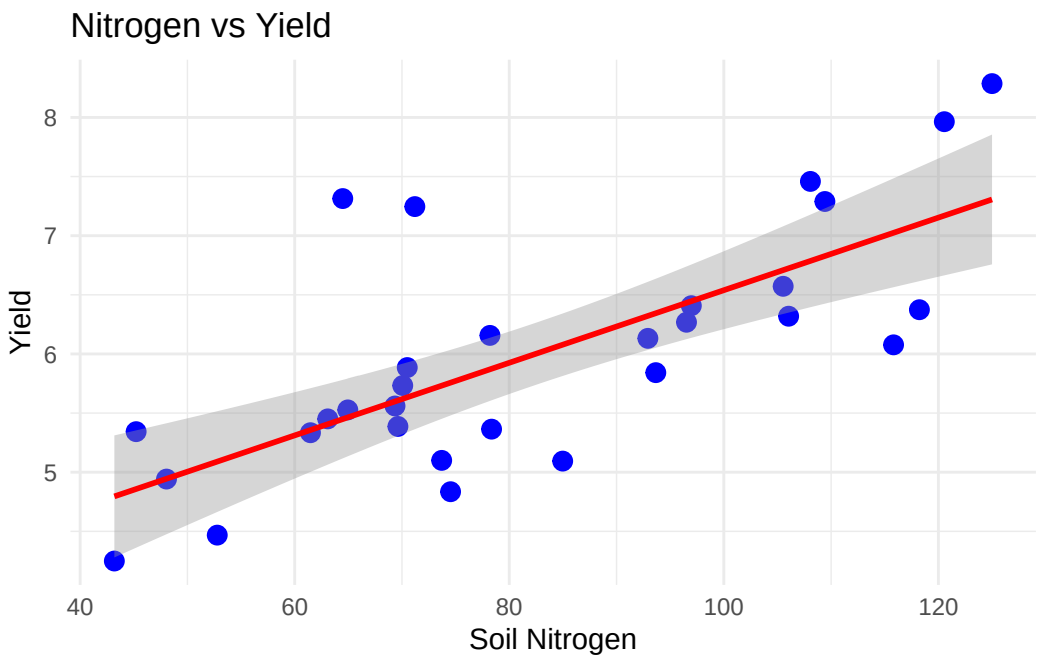
A plot shows what `summary()` numbers cannot: the **shape** of the relationship.

```
data2 <- read.csv("../data/raw/yield_soil_data.csv")
head(data2)
```

```
  nitrogen rainfall yield
1    70.49   530.07 5.885
2    73.70   517.01 5.100
3    96.97   655.49 6.408
4   109.42   573.61 7.289
5    63.08   613.72 5.451
6   120.55   736.67 7.964
```

7.1 Nitrogen vs yield

```
ggplot(data2, aes(nitrogen, yield)) +
  geom_point(color = "blue", size = 3) +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  labs(title = "Nitrogen vs Yield", x = "Soil Nitrogen", y = "Yield") +
  theme_minimal()
```



How to read it:

- **Linear trend?** If points follow a straight-line pattern, the relationship is linear.
- **Increasing yield?** If the line goes up, more nitrogen = higher yield.
- **Good fit?** If most points are close to the red line, the fit is strong.

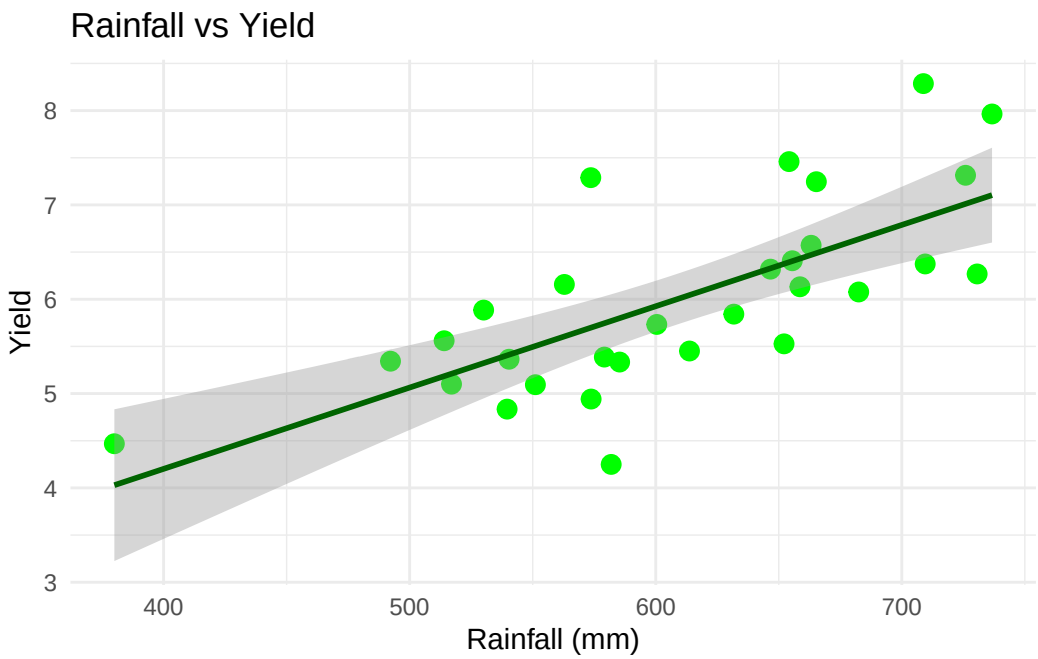
```
cor_n <- cor(data2$nitrogen, data2$yield)
round(cor_n, 3)
```

```
[1] 0.724
```

The correlation above puts a number on the same idea: closer to +1 means a tighter upward line.

7.2 Rainfall vs yield

```
ggplot(data2, aes(rainfall, yield)) +
  geom_point(color = "green", size = 3) +
  geom_smooth(method = "lm", se = TRUE, color = "darkgreen") +
  labs(title = "Rainfall vs Yield", x = "Rainfall (mm)", y = "Yield") +
  theme_minimal()
```



How to read it:

- You can see whether yield increases with rainfall.
- If the line is **flat**, rainfall might not impact yield much. If the line is **steep**, rainfall has a stronger effect.
- `se = TRUE` draws the **shaded area**: the uncertainty of the line. A narrow band means a confident line; a wide band means less certainty (usually at the ends where there is less data).

```
cor_r <- cor(data2$rainfall, data2$yield)
round(cor_r, 3)
```

```
[1] 0.709
```

8 Key Takeaways

1. Regression models a response (yield) from predictors (nitrogen, rainfall).
2. In `summary()`: **coefficients** give effects, **p-values** give importance, **RSE** gives typical error, **R^2 / adjusted R^2** give fit, **F-test** gives overall significance.
3. R^2 = SSR / SST, the share of variation explained. It is not "accuracy".
4. A high R^2 can hide overfitting; use **adjusted R^2** to compare models.
5. Missing factors (like irrigation) can bias predictions.

6. Always **plot** the data: the line and shaded band show trend, strength and uncertainty at a glance.